CSI-PAIRS: PAIRED ALIGNMENT AND INTERVENTION-RESPONSE SUPERVISION FOR MAP-CONDITIONED WIRELESS LOCALIZATION

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ABSTRACT

Map-conditioned wireless models are usually trained and evaluated on correctly matched signal–scene pairs. Such performance does not specify how their representations should change when local geometry changes while receiver state and radio configuration are held fixed. We formulate a paired audit that separates physically effective (active), ambiguous (gray), and below-noise (null) map interventions, and introduce CSI-PAIRS, a supervision framework on reusable scene-level world banks. For each Hamming-1 scene edit, Alignment asks which adjacent map better explains the observed channel, whereas Response predicts the ray-traced target channel under a signed edit. Both objectives update one retained map-conditioned state encoder; Response uses the clean, pair-consistent physical CSI patch as its primary target and a frozen-teacher latent patch as an auxiliary target, while the CSI-only teacher audits route sensitivity, and the headline model never receives receiver position. We specify a strict 2×2 experiment, scene-clustered statistics, shortcut controls, and claim-evidence gates that distinguish local paired sensitivity from global map validity, joint supervision from extra compute, and simulator-defined effects from real-world causality. A decisive no-position Response qualification compares oracle-x, no-x, copy, no-action, action-swap, map/edit-only, and null behavior before the protocol advances to two-city localization and external validation.

1 INTRODUCTION

Channel state information (CSI) contains a structured record of propagation through the physical environment. Recent channel foundation models learn from masked CSI reconstruction or multiple deterministic views of the same channel (Jiang et al., 2026a; 2025; 2026b), while multimodal systems combine signals with scene geometry or motion (Chen et al., 2026; Chu et al., 2026). These objectives can yield useful matched-pair representations. They do not, by themselves, identify how a representation should respond to an intervention on the scene: two predictors can agree on every observed zero-edit example and disagree arbitrarily after a wall is inserted, removed, or changed.

The distinction matters for localization. A map-conditioned predictor can use coarse city context, scene identity, or map artifacts without encoding the local geometry responsible for a particular channel. Conversely, degradation under an arbitrary wrong map does not prove physical grounding: the corruption can lie far outside the training distribution, while some local edits have no measurable effect at the receiver. A useful audit must therefore hold the receiver state, radio configuration, and propagation nuisance variables fixed; retrace the edited world; and avoid labeling every alternate map as physically incompatible.

We study this problem with *paired scene interventions*. A scene-level world bank contains canonical states of a small edit hypercube and is reused across many receiver positions. Only adjacent Hamming-1 worlds are compared. Each edge-position unit is routed by both a full-channel physical distance and a frozen-teacher distance. Active edges support relative Alignment; gray edges carry target regression but no strong compatibility label; null edges receive a tolerance constraint rather than a negative label.

PLANNED EXPERIMENT PANEL

Paired intervention audit. Replace with six-condition results for at least two C1-eligible external models: active and null paired alternatives, wrong city, geometry-destroyed, and empty map. Show paired bank-level intervals.

Figure 1: Planned paired-intervention audit. Panels report model information budgets, independent scene banks, seeds, paired intervals, and frozen active/null thresholds.

CSI-PAIRS adds two distinct constraints to a shared state encoder. Alignment selects the generating side of an active paired edge. Response predicts the direction and magnitude of the retraced raw-CSI target under a signed edit. The target is loaded only after source inputs are fixed; the route teacher reads CSI only, preventing the target map or receiver coordinate from leaking the answer. At localization time, only the retained state encoder and position head remain. The method therefore succeeds only if both skills are present in the shared representation, not merely in disposable audit heads.

This evidence-gated V2.1 draft makes three methodological contributions:

- We define a six-condition wrong-map audit and a paired compatibility protocol that distinguish active, gray, and null edits from arbitrary wrong-city, geometry-destroyed, and empty-map stress tests.
- We formulate node-balanced, bidirectional scene-level world banks and shared Alignment/Response supervision with a physical CSI target, an auxiliary CSI-only teacher target, no receiver position in the headline input, and explicit null dead zones.
- We preregister a strict 2×2 comparison, equal-resource and shortcut controls, scene-level inference, and downgrade rules. These tests, rather than the objective alone, determine whether joint value, cross-city localization, or bounded paired-context risk become eligible.

The central empirical question is whether no-position Response clears its information test and whether the two supervision factors then improve the retained localization representation beyond single-task and resource-matched controls. The gate sequence makes that question testable without tuning on target-city outcomes.

2 Related Work

Wireless self-supervision and world models. CSI-MAE masks and reconstructs CSI to learn channel representations (Jiang et al., 2026a). CSI-CLIP and CSI-CLIP++ align frequency-domain CSI with its delay-domain CIR view (Jiang et al., 2025; 2026b). The Wireless World Model combines CSI, point clouds, and trajectories through same-world predictive learning (Chen et al., 2026). These are strong starting points, but same-world completion and deterministic transforms of one channel do not supervise the response to a retraced scene edit. When source code is unavailable, our protocol labels the corresponding control "WWM-inspired same-world matched prediction" to distinguish it from a faithful reproduction.

Map-conditioned localization. SigMap uses a geographic map as a prompt for cross-scenario wireless localization (Chu et al., 2026). This makes it a direct diagnostic neighbor: our question is not whether a map improves matched localization, but whether the model responds correctly to local paired edits and remains stable on null edits. We evaluate external methods only under accurately named implementation status—official-code adaptation, paper-spec controlled implementation, or inspired control—and do not transfer numbers across datasets.

Geometry-aware wireless prediction. Wi-GATr predicts wireless quantities from scene primitives and transmitter/ receiver geometry (Hehn et al., 2025); WiSER predicts radiomaps and CIRs from a transmitter-conditioned scene memory (Qiao et al., 2026); RF inverse rendering explicitly models editable geometry and materials (Wang et al., 2026). These works motivate geometry-aware forward modeling. Their information budget differs from our headline setting: receiver position is excluded from CSI-PAIRS and appears only in a privileged oracle. Our contribution centers on paired supervision and an evidence protocol for the retained localization representation, rather than a new ray tracer or ranking loss.

3 Paired Problem Setup

3.1 Simulator-defined physical units

Let b index an independent scene tile, u a world state, x a receiver position, c the radio configuration, and ξ_{phys} all propagation nuisance variables that must remain shared across a pair. A deterministic ray-tracing backend produces the noise-free channel

$$H^{\star}_{b,u}(x, c, \xi_{\text{phys}}) = \mathcal{R}(M_{b,u}, x, c, \xi_{\text{phys}}). \tag{1}$$

An observed CSI sample adds an independently seeded observation process. For an intervention $u \leftrightarrow v$, the two calls share $(b, x, c, \xi_{\text{phys}})$ and differ only in one registered scene primitive. This is a simulator-defined paired unit. It does not establish a real-world counterfactual without an external engine or controlled measurement.

3.2 Reusable scene-level world banks

Each bank has $K = 2^d$ canonical worlds indexed by $u \in \{0,1\}^d$. A bit selects one of two physically admissible states for a registered primitive. Every world is rendered from its complete state description, never by replaying an edit history. The bank reuses one common receiver set X_b across all worlds, and all siblings remain in the same split. Training uses only

$$E_b = \{\{u, v\} : \text{Ham}(u, v) = 1\}, \tag{2}$$

with both directions sampled equally. Sampling is uniform over bank, node, position, and response query in that order. This construction removes the direct shortcut in which a per-position map uniquely identifies x, while unseen-bank, map-only, scene-ID, and variant-ID controls test residual shortcuts.

3.3 Effect routing

Let $\Phi_A(H)$ be the normalized full-channel physical representation, $\Phi_R^q(H)$ its query-patch restriction, and let $z_A = T_A(H)$ and $z_R^q = T_R^q(H)$ be the corresponding frozen CSI-only teacher representations. Source-only constants normalize the two pairs of symmetric distances

$$\begin{aligned} \delta^A_{uv} &= d_{\text{phys}}(\Phi_A(H_u), \Phi_A(H_v)), \quad & \gamma^A_{uv} &= d_z(z_{A,u}, z_{A,v}), \\ \delta^{R,q}_{uv} &= d_{\text{phys}}(\Phi^q_R(H_u), \Phi^q_R(H_v)), \quad & \gamma^{R,q}_{uv} &= d_z(z^q_{R,u}, z^q_{R,v}). \end{aligned} \tag{3}$$

Thresholds come from source repeat noise and a source-only practical-effect floor, then remain frozen:

$$\text{Route}(\delta, \gamma) = \begin{cases} \text{null}, & \delta \le \epsilon_0 \ \wedge \ \gamma \le \tau_0, \\ \text{active}, & \delta \ge \epsilon_1 \ \wedge \ \gamma \ge \tau_1, \\ \text{gray}, & \text{otherwise}. \end{cases} \tag{4}$$

Alignment uses $\text{Route}_A(u, v) = \text{Route}(\delta^A_{uv}, \gamma^A_{uv})$; Response uses the separate query-level route $\text{Route}_R(u, v, q) = \text{Route}(\delta^{R,q}_{uv}, \gamma^{R,q}_{uv})$, with separately frozen Alignment and Response thresholds. Thus a full-channel active edge need not make every Response query active. Route labels control losses and audit strata but are never model inputs. We report raw counts and scene coverage, physical-active/teacher-sensitive agreement, and physical-null/teacher-null agreement. Empty conditional means are qualification failures, not zeros.

3.4 What compatibility means

For an active edge $\{u, v\}$ at a fixed physical unit, the matched pairs are $\{(H_u, M_u), (H_v, M_v)\}$ and paired alternatives are $\{(H_u, M_v), (H_v, M_u)\}$. CSI and map marginals are identical between labels. A fixed-budget probe score on this support defines *compatibility decodability*, not global physical validity. Gray edges do not enter active AUROC. Null edges are evaluated for over-discrimination, not treated as negatives. Wrong-city, geometry-destroyed, and empty maps are stress tests and receive no calibrated probability unless their corruption families were included in source-only calibration.

PLANNED EXPERIMENT PANEL

Architecture. Draw source CSI/mask and source map entering shared F_θ; zero-action endpoint/Alignment and signed-edit Response entering one predictor; CSI-only target labels; discard all auxiliary heads before the common localization probe. Mark receiver x as oracle-only.

Figure 2: Planned CSI-PAIRS architecture. This is a method schematic, not evidence of efficacy. The final diagram must expose the input allowlist, stop-gradient target path, and the modules retained downstream.

4 CSI-PAIRS

4.1 Shared retained state

For CSI from world i, a supplied candidate map from world j, and mask m, define the retained state

$$C^m_{i|j} = F_\theta(V_m(H_i), G(M_j), c, m)\,, \tag{5}$$

with $C^m_u := C^m_{u|u}$. Here V_m exposes only unmasked source CSI, G is a map encoder, and the fusion module is shared by every arm. A predictor P_ψ receives (C^m_u, e_{uv}, q), where e_{uv} encodes the signed primitive edit and q is a target query. Zero action gives an identity prediction. The V2.1 primary target $y^q_v = \Psi(H_v)^q$ is the source-normalized clean raw-CSI patch under the pair-consistent gauge. Concretely, every sibling world at fixed (b, x) is rendered under the same registered complex reference $r_{b,x}$; Ψ applies that shared reference, the frozen antenna/subcarrier ordering and patching, and source-training mean/scale statistics. It performs no per-world phase fitting. The auxiliary target $t^q_v = T^q_R(H_v)$ is stop-gradient and frozen. The physical target is primary; the teacher target audits and regularizes the retained state but cannot replace physical success. Target CSI, position, route, and labels cannot enter a source predictor.

After pretraining, action encoder, predictor, teacher, and audit readouts are discarded. Localization uses only

$$r = F_\theta(H, M, c), \qquad \widehat{x} = g(r), \tag{6}$$

where g is trained with the same source-position labels and update budget for all arms. Thus any downstream difference must pass through the shared state.

4.2 Common endpoint objective

All four arms first learn the same matched endpoint objective. For a frozen query bank $Q(m)$ with $q \in m$,

$$\mathcal{L}_B = \mathbb{E}_{b,u,x,m,q}\,\rho_y(P_\psi(C^m_u, e_{uu}, q), y^q_u)\,. \tag{7}$$

Any common raw-CSI reconstruction term is identical across arms. Endpoint is therefore both a useful initialization and the 00 arm in the factorial test; it is not evidence of intervention response.

4.3 Paired Alignment

The same predictor defines an intrinsic energy

$$s_{\theta,\psi}(H_u, M_v) = -\frac{1}{|Q|}\sum_{q\in Q} \rho_{\mathrm{end}}\big(P_\psi(C_{u|v}, e_{vv}, q), (t^q_u, y^q_u)\big)\,, \tag{8}$$

where $\rho_{\mathrm{end}} = \rho_z + \lambda_y \rho_y$ is frozen across arms and the candidate map enters only through the supplied-map state $C_{u|v}$. On an active undirected edge, the symmetric margin loss is

$$\ell^{uv}_A = [m_{uv} - s_{\theta,\psi}(H_u, M_u) + s_{\theta,\psi}(H_u, M_v)]_+ \tag{9}$$

$$\quad + [m_{uv} - s_{\theta,\psi}(H_v, M_v) + s_{\theta,\psi}(H_v, M_u)]_+. \tag{10}$$

The margin is fixed or a source-frozen monotone function of physical effect. For null edges we instead penalize score gaps beyond a no-edit tolerance κ_s; gray edges receive no Alignment loss. Expectations first average within eligible banks and then across banks so route prevalence cannot silently reweight scenes.

4.4 INTERVENTION RESPONSE

For every directed edge, including active, gray, and null, the predictor emits $(\widehat{t}, \widehat{y})$ and regresses the retraced physical target plus the frozen auxiliary latent target:

$$\ell_{\text{target}}^{uv,m,q} = \rho_z\left(\widehat{t}_{u\to v}^{m,q}, t_v^q\right) + \lambda_y \rho_y(\widehat{y}_{u\to v}^{m,q}, y_v^q)\,. \tag{11}$$

Let $\widehat{\Delta y}_{uv} = \widehat{y}_{u\to v} - \widehat{y}_{u\to u}$ and $\Delta y_{uv} = y_v - y_u$. Active edges add direction-and-magnitude regression

$$\ell_{\Delta}^{uv} = \rho_y(\widehat{\Delta y}_{uv}, \Delta y_{uv}). \tag{12}$$

Null edges still regress their exact target but penalize hallucinated changes outside source-frozen dead zones:

$$\ell_{0R}^{uv} = [\|\widehat{\Delta y}_{uv}\|_{\text{phys}} - \kappa_y]_+^2. \tag{13}$$

Gray edges carry only target regression. The target-world CSI is used to form frozen targets, physical labels, routes, and metrics; it never enters a source predictor.

4.5 FACTORIAL OBJECTIVE

Let $a, r \in \{0, 1\}$ enable Alignment and Response. Each arm optimizes

$$\mathcal{L}_{a,r} = \mathcal{L}_B + a\lambda_A\mathcal{L}_A + r\lambda_R\mathcal{L}_R. \tag{14}$$

The four arms share initialization, backbone, raw world banks, teacher, mask/query bank, source-only selection budget, downstream head, and random draws. Factor weights are normalized from a source-only pilot and remain the same in a single branch and Full. We report effective updates, gradient norms, parameters, FLOPs, and wall time. Full is a valid headline only if the two single branches first solve their own tasks, Full preserves both, and equal-resource/mechanical-concatenation controls do not explain downstream performance.

5 IDENTIFICATION AND TESTABLE IMPLICATIONS

Matched support leaves intervention response underdetermined. Correct-pair training identifies zero-action behavior. Any two predictors equal almost everywhere on that support have the same matched risk while allowing different extensions to nonzero action. Paired retraced targets supply the missing off-support constraint.

Symmetric quartets match unimodal marginals. For an active edge, matched and alternative quartets each contain $\{H_u, H_v\}$ and $\{M_u, M_v\}$. Therefore a CSI-only or map-only statistic cannot solve the label. Joint world-ID matching remains possible and must be tested empirically through bank reuse, unseen banks, and ID controls.

Response specifies more than rank. Zero active ranking loss requires matched scores to exceed alternatives by a margin. Exact identity and target regression additionally imply $\widehat{\Delta z}_{uv} = \Delta z_{uv}$. This is a directional and magnitude constraint; the preregistered factorial interaction measures whether the two tasks are synergistic.

Action and source-CSI value are empirical questions. For state S, edit A, and target Z, squared-loss Bayes risks satisfy

$$R_{\text{noA}}^* - R_A^* = \mathbb{E}[\operatorname{tr}\operatorname{Cov}(\mathbb{E}[Z \mid S, A] \mid S)] \geq 0. \tag{15}$$

The inequality is strict only if conditional targets vary with action. A learned action encoder may still collapse. Likewise adding visible source CSI cannot increase Bayes risk over map/edit-only input, but finite models may not use it. No-action, action-swap, map/edit-only, CSI-only, and privileged-x experiments are therefore required.

PLANNED EXPERIMENT PANEL

Split and gate flow. Show seven disjoint source roles: encoder-train, method-select, probe-train/select, calibration-fit/select, and final-unseen; then target support/query in two cities. Show G0 literature/resources, G1 world-bank/RT qualification, G2 teacher and No-X qualification, G3 single-branch, G4 seven-part joint-value, G5 two-city localization, G6 risk, G7 path-mechanism, and G8 external-validity exits.

Figure 3: Planned split, gate, and claim flow. The highest independent unit is the scene bank. Each downstream claim activates only after its upstream gate, with thresholds frozen before held-out target evaluation.

6 Experimental Protocol

6.1 Questions and independent units

The experiment answers five questions in order. RQ1 asks whether at least two C1-eligible map-conditioned models exhibit an active/null wrong-map pattern. RQ2 asks whether Alignment learns paired compatibility without null over-discrimination. RQ3 asks whether masked no-position Response predicts retraced targets better than copy, no-action, and action-swap in each held-out selection scene. RQ4, reached only after the earlier gates, asks whether Full improves localization in two unseen cities beyond both single branches and resource-matched controls. RQ5 asks whether, on a frozen paired-proposal set and strict $k = 0$ common support, separately calibrated localization-failure risk improves AURC over random and compatibility-only baselines while retaining registered calibration, coverage, and monotonicity properties.

The highest independent unit is the scene tile/world bank. All positions, worlds, edges, directions, masks, and augmentations from a bank remain in one cluster. We first compute within-bank numerators, denominators, and location summaries, then perform scene-macro aggregation and scene-cluster bootstrap. Position-level confidence intervals are descriptive only and never substitute for independent scenes.

6.2 Six-condition map audit

For fixed channel samples and denominators, each model receives: correct map, paired active alternative, paired null alternative, wrong-city map, geometry-destroyed map with preserved low-order statistics, and empty map. Primary localization outcomes are median and P90 error, reported per city and bank. Active compatibility uses a frozen, equal-budget probe AUROC. Gray score gaps and null over-discrimination are separate. The latter three arbitrary corruptions are stress tests, not paired causal evidence.

6.3 Strict controls

Table 1 fixes the minimum control suite. External baselines come after the internal four arms because different backbones cannot isolate the two supervision factors.

6.4 Metrics and gates

Alignment's primary diagnostic is active paired CGS AUROC under a unified probe; Response's primary diagnostic is active, full-channel, target-free physical-patch NMSE. Secondary Response metrics include latent NMSE, delta cosine, magnitude ratio, and null hallucination. Localization reports median/P90 at $k \in \{0, 8, 32, 128\}$ unique target positions, with support and query siblings isolated. The paired-proposal failure score reports AURC, ECE, Brier, and NLL only on frozen source-calibration support; arbitrary single-map deployment is outside this probability claim.

The gate order is: (i) each single branch beats its task control; (ii) Full is non-inferior on both native skills; (iii) Full improves preregistered $-\log$ bank-median localization utility over both single branches; (iv) the factorial interaction is positive; and (v) no target city or $k = 0/8$ setting has an excessive reverse effect. Multiplicity within a gate is controlled by Holm correction or simultaneous scene-cluster intervals. A failed upstream gate makes downstream tests exploratory.

Table 1: Required controls and the alternative explanation each tests.

Control	Purpose
Endpoint/A/R/Full	Isolate Alignment, Response, and their interaction
copy; no-action; action-swap	Test target predictability and use of signed edits
without-map; CSI-only	Test whether geometry enters Response
map-only; edit-only	Test average-edit and map lookup shortcuts
scene-ID; variant-ID	Detect world-bank identity leakage
oracle-x	Quantify the privileged-position information gap
shuffled pairing	Detect gains from generic extra supervision
equal-FLOP single branches	Separate task value from compute
parameter/FLOP-matched concat	Separate sharing from mechanical concatenation
full-width 2× concat	Generous two-encoder upper control

PLANNED EXPERIMENT PANEL

Formal result panel. A: external six-condition wrong-map. B: No-X vs copy/no-action/action-swap active NMSE. C: null violation rate with the frozen tolerance. D: Endpoint/A/R/Full localization at $k = 0, 8$ in both target cities. Optional E: preregistered interaction with cluster CI.

Figure 4: Planned formal results. Panels are generated from authenticated per-sample files and report scene banks, training seeds, label draws where applicable, units, confidence intervals, and frozen pass thresholds.

7 Experiment Readiness and Launch Rule

Table 2 separates completed software checks from the launch criteria for formal experiments. Synthetic qualification exercises file formats, samplers, allowlists, metrics, checkpoints, and stop decisions. Its archived no-position outcomes motivate a gate-first launch on independent ray-traced scene banks.

The experiment therefore starts with a qualified RT engine, independent assets, source-frozen physical and teacher targets, and the no-position gate on new scene banks. An oracle-x pass paired with a no-x failure identifies an information gap and triggers the preregistered Alignment-only fallback. A no-x pass over copy, no-action, and action-swap with null safety unlocks the four-arm experiment.

8 Limitations and Conclusion

Limitations. The primary evidence concerns paired interventions generated by the qualified simulator; G8 measures transfer to an independent engine. Physical targets and teacher/physical agreement address teacher blind spots, while the oracle-x gap measures ambiguity from hidden receiver position. Compatibility is scoped to the frozen paired proposal set. Scene-bank inference reflects the available independent sample count, and resource-matched controls separate supervision value from additional compute.

Ethics Statement

Wireless localization can expose sensitive movement and location information and can be used for surveillance. Any real measurements must follow consent, privacy, institutional, and data-governance requirements. Released artifacts must remove device identifiers, private coordinates, credentials, and proprietary infrastructure paths. Licenses and permitted uses for maps, simulator assets, datasets, and baseline code must be documented. The method is not a safety or deployment certification.

Conclusion. CSI-PAIRS turns a vague expectation—that a map-conditioned wireless model should react to geometry—into a paired, routed, and falsifiable supervision problem. Scene-level reuse restricts position lookup; symmetric active quartets support relative Alignment; physical and

Table 2: Formal experiment launch sequence. Software readiness and scientific gate completion are tracked separately.

Stage	Readiness	Launch criterion
Official format/policy	Ready	Preserve anonymity and the AI use statement
Canonical world-bank code	Ready	Pass independent RT regeneration
Six-condition pipeline	Ready	Execute two C1-eligible models
Teacher and routing	Ready	Pass per-bank teacher and route coverage
Masked no-x Response	Gate first	Beat three controls per scene and pass null safety
Strict four arms	Queued	Start after the Response stop/go review
Unseen-city localization	Queued	Start after both native skill gates
External validity	Queued	Test cross-engine transfer after internal gates

teacher targets define Response; and strict controls determine whether the information survives in the retained localization state. The implementation turns these requirements into executable gates. Its next milestone is the independent Response qualification that decides whether the full factorial and two-city localization study should proceed.

AI Use Statement

In this work, we used OpenAI Codex to assist with refining the conceptual framework and hypotheses, critiquing research methodology and experimental design, drafting and debugging software, searching and summarizing related literature, organizing the manuscript, and improving translation and readability. We treated its outputs as suggestions rather than evidence. AI-assisted material was checked, as applicable, against the frozen protocol, repository source, unit and mutation tests, original cited papers, and official ICLR policy pages. No scientific result was generated or accepted solely on the basis of an AI response. The authors take responsibility for the final text, claims, code, and artifacts.

Reproducibility Statement

The supplement packages the world-bank schema, frozen configuration, model allowlists, code-level semantic tests, run scripts, and claim-evidence contracts. A result-bearing release additionally includes the independent renderer adapter and regeneration gate, RT configuration and calibration report, asset-permission ledger, seven-way source split ledger, teacher/readout qualification, per-sample outputs, checkpoints, cluster-level sufficient statistics, external-baseline adaptation records, and signed SHA-256 manifests.

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A Proof Sketches and Boundaries

A.1 Same-world non-identification

Let $\mathcal{S}_0$ be the support of inputs with zero action. If predictors f and g agree almost everywhere on $\mathcal{S}_0$, then for any matched loss ℓ, their matched risks are equal by substitution. Define g to equal f on $\mathcal{S}_0$ and take any measurable value on disjoint nonzero action support $\mathcal{S}_1$. Matched risk cannot distinguish the extension. The result establishes only underdetermination off support.

A.2 Marginal equality of active quartets

For a fixed active edge, matched and alternative multisets are

$$\begin{aligned} Q_+ &= \{(H_u, M_u), (H_v, M_v)\}, \\ Q_- &= \{(H_u, M_v), (H_v, M_u)\}. \end{aligned}$$

Projection onto CSI gives $\{H_u, H_v\}$ for both; projection onto maps gives $\{M_u, M_v\}$ for both. Direction-symmetric routing and equal weighting preserve the equality. Hence a statistic of only one projection cannot distinguish labels. A joint identifier is not ruled out.

A.3 Bayes-risk value of action

For squared loss, the optimal predictors are conditional means. Applying the law of total covariance to $\mathbb{E}[Z \mid S, A]$ conditioned on S yields

$$R^*_{\mathrm{noA}} - R^*_A = \mathbb{E}\|\mathbb{E}[Z \mid S, A] - \mathbb{E}[Z \mid S]\|_2^2 = \mathbb{E}\,\mathrm{tr}\,\mathrm{Cov}(\mathbb{E}[Z \mid S, A] \mid S) \geq 0.$$

Strictness requires action-dependent conditional means on positive measure. It neither guarantees that a finite network reaches Bayes risk nor that a learned action embedding retains the raw action.

B Complete Sampling and Leakage Contract

For each optimization draw: sample an eligible bank uniformly; sample a source node uniformly; sample a common position uniformly without route conditioning; sample each legal Hamming-1 target branch with positive probability; and reuse the same observation augmentation, mask, and query across the branch bundle. Target CSI is loaded only after the source-predictor inputs are finalized. The data loader asserts the following no-position allowlist:

$$\{V_m(H_u), G(M_u), \Delta_{uv}, c, m, q\}.$$

It rejects x, target CSI, target latent before label construction, route, effect magnitude, bank/world ID, and any user-centered crop. The oracle-x loader is a separate code path and checkpoint.

Siblings remain clustered across source-encoder-train, source-method-selection, source-probe-train, source-probe-selection, source-calibration-fit, source-calibration-selection, source-final-unseen-bank, target adaptation support, and final target query. No partition can be renamed "pilot" to bypass this ledger.

Table 3: Alignment and Response result schema for formal scene-bank experiments.

Arm	Active CGS ↑	Null gap ↓	Response NMSE ↓	FLOPs
Endpoint	PLANNED	PLANNED	PLANNED	PLANNED
Alignment only	PLANNED	PLANNED	PLANNED	PLANNED
Response only	PLANNED	PLANNED	PLANNED	PLANNED
Full	PLANNED	PLANNED	PLANNED	PLANNED

Table 4: Unseen-city localization and paired-risk result schema. Risk columns are permitted only for the frozen $k = 0$ paired-proposal audit.

City	k	Median ↓	P90 ↓	AURC ↓	ECE ↓
City A	0/8/32/128	PLANNED	PLANNED	PLANNED	PLANNED
City B	0/8/32/128	PLANNED	PLANNED	PLANNED	PLANNED

C Planned Main Tables

D Artifact-to-Claim Checklist

Before replacing any dash, the artifact must include scene IDs, split role, route counts, sample denominator, seed, model/input allowlist, checkpoint hash, configuration hash, sufficient statistics, and failure cases. A result enters the abstract only after its corresponding gate passes. Specifically: a wrong-map domain claim needs at least two runnable map-conditioned models; Response needs teacher agreement and per-scene wins over copy, no-action, and action-swap; joint value needs both native skills, equal-resource controls, and a positive preregistered interaction; cross-city claims need both target cities; and any simulator-consistent wording needs qualified RT plus an external engine or controlled real intervention.